Project Report — UAV Networks: Access Point Scenario (Case 2)

A classical-ML intrusion detection system (IDS) on the UAV-NIDD dataset, Access Point scenario. Built in six stages; every number below comes from a real, reproducible run (seed 42).

1. Dataset and scenario

UAV-NIDD (Hadi et al., IEEE TNSE 2025, Zenodo 15125851) is a real UAV testbed capture with three compromise scenarios. This project uses Case 2: a compromised Access Point that can intercept, redirect or observe traffic between the UAVs and the ground station — especially relevant to Man-in-the-Middle and Evil Twin attacks.

- Packet-level Wi-Fi capture: radiotap, 802.11, IP, UDP and HTTP fields.
- 631,356 packets, 50 features, 10 attack classes + Normal.
- Imbalance ≈ 4,300 : 1 (Brute-Force 172,509 vs Reconnaissance 40).

2. Data-quality problems found

- The label column's header is 'Normal' (the real header was lost) → renamed 'Class'.
- 'Ewil Twin' typo → 'Evil Twin'.
- Column headers are shifted against their contents (e.g. frame.time_relative holds the epoch time, radiotap.datarate holds the channel frequency, ip.proto holds IP addresses).
- Cells mix numbers, hex and text. Hex is parsed; other text becomes -1. Rows are kept: dropping them would delete 100% of Evil Twin, MITM, Reconnaissance and Replay.

3. Pipeline and the two paths

Load → clean → split 80/20 stratified → PCC |corr| > 0.85 → top-25 features → RandomUnderSampler + Borderline-SMOTE → model → macro-F1. Everything that learns from data is fit on the training set only.

- report path: exactly as the GP report says (all columns kept, ANOVA top-25).
- search path: honest version — drops 4 time/order and 4 IP columns, RF-importance top-25.

4. Results

Stage 1 — shortcut check

A decision tree using one column only: frame.time_relative (really the capture timestamp) reaches macro-F1 0.923 on its own. Each attack was captured in its own session, so time reveals the class (data leakage).

column	macro_f1_alone	dropped_in_search
frame.time_relative	0.923	True
radiotap.channel.flags.cck	0.614	True
frame.len	0.422	False
frame.time_epoch	0.349	False
radiotap.rxflags	0.344	True
frame.encap_type	0.330	False

Stage 2 — class balancing (test macro-F1)

path	strategy	macro_f1	weighted_f1	accuracy
report	none	0.9586	0.9620	0.9619
report	under_smote	0.9691	0.9751	0.9737
report	class_weight	0.9595	0.9624	0.9619
search	none	0.8860	0.9494	0.9530
search	under_smote	0.8853	0.9481	0.9492
search	class_weight	0.8834	0.9442	0.9449

The report's claimed +0.18 from balancing does not reproduce (all within 0.01). under_smote was kept: it lifts the recall of DDoS, MITM and Replay.

Stage 3 — binary, Normal vs Attack (5-fold CV)

path	cv_macroF1	cv_normal_recall	cv_normal_prec	cv_MCC	baseline_allAttack_macroF1
report	1.0000	1.0000	1.0000	1.0000	0.4428
search	1.0000	1.0000	1.0000	1.0000	0.4428

1.000 even on the honest path is a red flag. Every Normal packet has frame.encap_type = 23 and no attack packet does: Normal was captured with a different setup. Dropping the strongest columns barely helps:

columns_dropped	which	test_macroF1
0	-	1.0000
1	frame.encap_type	0.9999
4	frame.encap_type, arp, eapol.keydes.replay_counter, wlan_radio.data_rate	0.9998
6	frame.encap_type, arp, eapol.keydes.replay_counter, wlan_radio.data_rate, frame.len,	0.9299

	udp.dstport	

Stage 4 — multi-class, 11 classes (5-fold CV)

path	cv_macroF1	cv_weightedF1
report	0.9742	0.9751
search	0.8846	0.9492

class	support	report_f1	search_f1
Brute-Force	138007	1.000	0.999
DDoS	25289	0.771	0.549
De-authentication	124451	0.945	0.923
DoS	18890	1.000	1.000
Evil Twin	8027	1.000	0.850
Fake Landing	19667	1.000	0.994
MITM	4013	1.000	0.718
Normal	103633	1.000	1.000
Reconnaissance	32	1.000	1.000
Replay	6820	1.000	0.697
Scanning	56255	1.000	1.000

Weak classes on the honest path: DDoS, Replay, MITM. Reconnaissance (32 training rows) is not reliable.

Stage 5 — model benchmark (test macro-F1)

model	report	search
XGBoost	0.9795	0.8975
RandomForest	0.9691	0.8853
DecisionTree	0.9747	0.8723
LogisticRegression	0.9092	0.6828
LinearSVM	0.9059	0.6801

Tree models clearly beat linear models. Best honest model: XGBoost (0.8975). The 0.90 target is met only on the leaky report path.

Stage 6 — importance, cost, reproducibility

Permutation importance, top 5 — report path: radiotap.rxflags, frame.time_relative, frame.number, wlan.seq, wlan.fc.moredata (time/order columns lead). Search path: frame.encap_type, wlan_radio.data_rate, udp.dstport, data.len, wlan.seq.

path	train (s)	inference (s, 126272 rows)	rows/sec	model size (MB)	reproducible
report	2.5000	0.1700	722,477	134.0000	True
search	2.4000	0.1900	676,996	227.0000	True

5. Conclusion

- Report-style results (~0.97 macro-F1) rely on capture timestamps.
- Trustworthy multi-class result: 0.885 (RF, CV) / 0.897 (XGBoost, test).
- Binary detection on AP is not a fair test: Normal was captured with a different setup.
- Next step: a time-aware split (train on earlier packets, test on later ones).